

Algebra Sparks

Every graded set, every week — 4 weeks, 20 two-page sheets, with an answer key after each week.

WEEK 1

5 sets

WEEK 2

5 sets

WEEK 3

5 sets

WEEK 4

5 sets

HOW TO USE IT

One page per day. Warm-Up is recall, Core is the graded tier, Challenge is never graded — the score box counts the graded items only. The same items appear in the app's Practice Bay, so a day can be done on paper or on screen, not both.

Find the Rule

Say the rule out loud in one sentence before you write the next number.

Warm-Up 10 ITEMS • RECALL

1 3, 6, 9, 12 — next

2 5, 10, 15, 20 — next

3 2, 4, 8, 16 — next

4 20, 17, 14, 11 — next

5 1, 4, 7, 10 — next

6 100, 90, 80 — next

7 2, 4, 6, 8 — next

8 10, 20, 30 — next

9 1, 3, 5, 7 — next

10 4, 8, 12 — next

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

11. Rule $\times 3$, starting at 2 — the fourth number (2, 6, 18, 54)

.....

12. 1, 3, 7, 15 — next (Double it and add one)

.....

13. Rule $\times 2 + 1$. Put in 5

.....

14. 2, 5, 10, 17 — next (The gaps go 3, 5, 7, 9)

.....

15. 7, 14, 21, 28 — next

.....

16. 1, 4, 9, 16 — next

.....

17. 3, 9, 27 — next

.....

Function Machines

In goes a number, out comes a number. The machine never changes its mind halfway.

Warm-Up 12 ITEMS • RECALL

1 Machine + 8. In 5

2 Machine $\times$ 6. In 4

3 Machine $-$ 7. In 20

4 Machine $\times$ 10. In 9

5 Machine $\div$ 2. In 18

6 Machine + 15. In 15

7 Machine $\times$ 2. In 5

8 Machine $\times$ 3. In 4

9 Machine + 7. In 8

10 Machine $-$ 3. In 10

11 Machine $\times$ 10. In 6

12 Machine $\div$ 2. In 14

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. Machine $\times$ 3 + 4. In 6 (Multiply first)

.....

14. Machine $\times$ 5 $-$ 2. In 7

.....

15. Machine $\times$ 2 + 9. In 12

.....

16. Machine $\times$ 3 + 4. Out 25 — what went in (Undo the + 4 first)

.....

17. Machine $\times$ 4 $-$ 1. Out 27 — what went in

.....

18. Machine $\times$ 3 + 1. In 7

.....

19. Machine $\times$ 4 $-$ 2. In 6

.....

20. Machine $\times$ 5 + 3. In 9

.....

21. Machine $\times$ 2 + 10. In 12

.....

22. Machine $\times$ 6 $-$ 5. In 8

.....

★ Challenge NOT GRADED • REASON IT OUT

23. In 2 gives 7, in 3 gives 10, in 4 gives 13. In 10 gives (The rule is $\times 3 + 1$)

24. In 1 gives 5, in 2 gives 9, in 3 gives 13. In 12 gives (The rule is $\times 4 + 1$)

25. Machine $\times 3 + 4$ gives 31 — what went in

26. Machine $\times 7 - 2$ gives 47 — what went in

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Letters Standing for Numbers

A letter is a number nobody has named yet. Put the number in, then work it out.

Warm-Up 12 ITEMS • RECALL

1 $n = 4$. Value of $n + 6$

2 $n = 4$. Value of $3n$

3 $n = 7$. Value of $n - 3$

4 $n = 5$. Value of $2n$

5 $n = 9$. Value of $n + n$

6 $n = 6$. Value of $10 - n$

7 $n = 3$. Value of $n + 5$

8 $n = 3$. Value of $4n$

9 $n = 8$. Value of $n - 5$

10 $n = 6$. Value of $2n$

11 $n = 7$. Value of $n + n$

12 $n = 4$. Value of $12 - n$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. $n = 5$. Value of $3n + 2$

.....

14. $n = 8$. Value of $2n - 5$

.....

15. $a = 3$, $b = 7$. Value of $4a + b$ (12 and 7)

.....

16. $n = 6$. Value of $n \times n$

.....

17. $n = 12$. Value of $n \div 4 + 5$

.....

18. $n = 6$. Value of $3n + 4$

.....

19. $n = 9$. Value of $2n - 7$

.....

20. $n = 5$. Value of $n \times n$

.....

21. $n = 20$. Value of $n \div 4 + 3$

.....

22. $n = 11$. Value of $4n - 14$

.....



Working space



Error journal

One-Step Equations

Do the same thing to both sides. Type the value of the letter.

Warm-Up 12 ITEMS • RECALL

1 $x + 4 = 11$

2 $x - 5 = 9$

3 $3x = 21$

4 $x + 12 = 20$

5 $6x = 30$

6 $x \div 2 = 8$

7 $x + 3 = 10$

8 $x + 6 = 14$

9 $x + 9 = 15$

10 $x + 4 = 4$

11 $x + 11 = 20$

12 $x + 2 = 13$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. $x + 17 = 42$

14. $8x = 96$ ($96 \div 8$)

15. $x - 23 = 19$

16. $x \div 5 = 13$

17. $4x = 100$

18. $x + 24 = 51$

19. $x - 7 = 12$

20. $x - 31 = 24$

21. $x + 47 = 100$

22. $x - 18 = 33$



Working space



Error journal

Guess My Rule & the Math Trail

Capstone day. Every stop on the trail is a problem you wrote yourself.

Warm-Up 4 ITEMS • RECALL

1 Stops on a ten-stop trail

2 $x + 9 = 9$

3 $3x = 12$

4 $5x = 20$

Core GRADED • SHOW YOUR WORKING

STOP RULE Short Core today — do all of them, then turn the page over.

5. Trail rule $\times 2 + 3$. Stop number 6 gives

.....

6. Ten stops, 3 minutes each — total minutes

.....

7. $9x = 72$

.....

8. $x \div 4 = 11$

.....

Challenge

9. In 5 gives 23, in 8 gives 35. In 10 gives (The rule is $\times 4 + 3$)

10. $2x + 2x = 40$ (That's $4x$)

11. A rule turns 4 into 4 and 6 into 8. Put in 10 ($\times 2 - 4$)

12. $11x = 253$

13. $2x + 7 = 25$

14. $3x - 5 = 22$



Working space



Error journal

Week 1 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

1.1 • Find the Rule

OUT OF 17

Warm-Up • 1-10 • 15 | 25 | 32 | 8 | 13 | 70 | 10 | 40 | 9 | 16

Core • 11-17 • 54 | 31 | 11 | 26 | 35 | 25 | 81

Challenge • 18-22 • 49 | 25 | 101 | 13 | 42

1.2 • Function Machines

OUT OF 22

Warm-Up • 1-12 • 13 | 24 | 13 | 90 | 9 | 30 | 10 | 12 | 15 | 7 | 60 | 7

Core • 13-22 • 22 | 33 | 33 | 7 | 7 | 22 | 22 | 48 | 34 | 43

Challenge • 23-26 • 31 | 49 | 9 | 7

1.3 • Letters Standing for Numbers

OUT OF 22

Warm-Up • 1-12 • 10 | 12 | 4 | 10 | 18 | 4 | 8 | 12 | 3 | 12 | 14 | 8

Core • 13-22 • 17 | 11 | 19 | 36 | 8 | 22 | 11 | 25 | 8 | 30

Challenge • 23-26 • 20 | 28 | 36 | 37

1.4 • One-Step Equations

OUT OF 22

Warm-Up • 1-12 • 7 | 14 | 7 | 8 | 5 | 16 | 7 | 8 | 6 | 0 | 9 | 11

Core • 13-22 • 25 | 12 | 42 | 65 | 25 | 27 | 19 | 55 | 53 | 51

Challenge • 23-26 • 7 | 11 | 266 | 200

Fri • Guess My Rule & the Math Trail

OUT OF 8

Warm-Up • 1-4 • 10 | 0 | 4 | 4

Core • 5-8 • 15 | 30 | 8 | 44

Challenge • 9-14 • 43 | 10 | 16 | 23 | 9 | 9

What a Letter Is

n is a seat a number sits in. $3n$ is three of them.

Warm-Up 12 ITEMS • RECALL

1 $n = 4$. Value of $n + 6$

2 $n = 4$. Value of $3n$

3 $n = 7$. Value of $n - 3$

4 $n = 5$. Value of $2n$

5 $n = 9$. Value of $n + n$

6 $n = 6$. Value of $10 - n$

7 $n = 5$. Value of $n + 8$

8 $n = 5$. Value of $4n$

9 $n = 9$. Value of $n - 4$

10 $n = 7$. Value of $3n$

11 $n = 8$. Value of $n + n$

12 $n = 3$. Value of $15 - n$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. $n = 5$. Value of $3n + 2$

.....

14. $n = 8$. Value of $2n - 5$

.....

15. $n = 6$. Value of $n \times n$

.....

16. $n = 12$. Value of $n \div 4 + 5$

.....

17. $n = 10$. Value of $4n - 10$

.....

18. $n = 7$. Value of $4n + 5$

.....

19. $n = 9$. Value of $3n - 8$

.....

20. $n = 8$. Value of $n \times n$

.....

21. $n = 24$. Value of $n \div 6 + 7$

.....

22. $n = 15$. Value of $3n - 20$

.....



Working space



Error journal

Evaluate It

Substitute the number, then work it out. One step at a time.

Warm-Up 12 ITEMS • RECALL

1 $n = 2$. Value of $5n$

2 $n = 3$. Value of $n + 7$

3 $n = 10$. Value of $n \div 2$

4 $n = 1$. Value of $9n$

5 $n = 0$. Value of $6n$

6 $n = 8$. Value of $n - 8$

7 $n = 3$. Value of $7n$

8 $n = 4$. Value of $n + 11$

9 $n = 12$. Value of $n \div 3$

10 $n = 2$. Value of $8n$

11 $n = 0$. Value of $9n$

12 $n = 6$. Value of $n - 6$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. $n = 7$. Value of $4n + 1$

.....

14. $n = 6$. Value of $5n - 12$

.....

15. $n = 9$. Value of $2n + 2$

.....

16. $n = 11$. Value of $3n - 3$

.....

17. $n = 4$. Value of $25 - 3n$

.....

18. $n = 8$. Value of $5n + 2$

.....

19. $n = 7$. Value of $6n - 10$

.....

20. $n = 10$. Value of $3n + 5$

.....

21. $n = 12$. Value of $4n - 8$

.....

22. $n = 5$. Value of $40 - 4n$

.....

Write the Expression

Turn the sentence into symbols. Then check it on a number.

Warm-Up 12 ITEMS • RECALL

1 Three lots of n , when $n = 4$

2 n more than 5, when $n = 3$

3 Double n , when $n = 7$

4 n less than 10, when $n = 4$

5 Half of n , when $n = 8$

6 n plus n , when $n = 5$

7 Four lots of n , when $n = 5$

8 n more than 8, when $n = 4$

9 Double n , when $n = 9$

10 n less than 20, when $n = 7$

11 Half of n , when $n = 12$

12 n plus n , when $n = 11$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Five more than double n , when $n = 6$

14. Three less than four lots of n , when $n = 5$

15. Ten more than half of n , when $n = 20$

16. Triple n , plus 4, when $n = 8$

17. n multiplied by itself, when $n = 7$

18. Seven more than double n , when $n = 8$

19. Four less than five lots of n , when $n = 6$

20. Twelve more than half of n , when $n = 30$

21. Quadruple n , plus 6, when $n = 7$

22. n multiplied by itself, when $n = 9$



Working space



Error journal

Two Letters

Substitute both before you simplify anything.

Warm-Up 12 ITEMS • RECALL

1 $a = 2, b = 3$. Value of $a + b$

2 $a = 2, b = 3$. Value of ab

3 $a = 5, b = 1$. Value of $a - b$

4 $a = 4, b = 4$. Value of $a + b$

5 $a = 3, b = 6$. Value of $b \div a$

6 $a = 1, b = 9$. Value of ab

7 $a = 3, b = 4$. Value of $a + b$

8 $a = 3, b = 4$. Value of ab

9 $a = 9, b = 2$. Value of $a - b$

10 $a = 5, b = 5$. Value of $a + b$

11 $a = 4, b = 12$. Value of $b \div a$

12 $a = 1, b = 11$. Value of ab

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. $a = 3, b = 7$. Value of $4a + b$

.....

14. $a = 4, b = 6$. Value of $3a + 2b$

.....

15. $a = 5, b = 2$. Value of $a \times b + 3$

.....

16. $a = 6, b = 3$. Value of $2a - b$

.....

17. $a = 8, b = 2$. Value of $a \div b + 5$

.....

18. $a = 5, b = 6$. Value of $4a + b$

.....

19. $a = 3, b = 8$. Value of $3a + 2b$

.....

20. $a = 7, b = 3$. Value of $a \times b + 4$

.....

21. $a = 9, b = 5$. Value of $2a - b$

.....

22. $a = 12, b = 3$. Value of $a \div b + 6$

.....

Perimeter as a Rule

Write it as $4s$ once, then test it on three squares.

Warm-Up 4 ITEMS • RECALL

1 Perimeter of a square of side 3

2 Of side 5

3 Perimeter of a square of side 4

4 Of side 8

Core GRADED • SHOW YOUR WORKING

STOP RULE Short Core today — do all of them, then turn the page over.

5. Using $4s$ with $s = 9$ — the perimeter

.....

6. A square of perimeter 28 — its side

.....

7. Using $4s$ with $s = 11$ — the perimeter

.....

8. A square of perimeter 44 — its side

.....

14. Using 4s, the side when the perimeter is 144

Week 2 Answers

Mark the reasoning, not the handwriting.

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

2.1 • What a Letter Is

OUT OF 22

Warm-Up • 1-12 • 10 | 12 | 4 | 10 | 18 | 4 | 13 | 20 | 5 | 21 | 16 | 12

Core • 13-22 • 17 | 11 | 36 | 8 | 30 | 33 | 19 | 64 | 11 | 25

Challenge • 23-26 • 20 | 28 | 35 | 24

2.2 • Evaluate It

OUT OF 22

Warm-Up • 1-12 • 10 | 10 | 5 | 9 | 0 | 0 | 21 | 15 | 4 | 16 | 0 | 0

Core • 13-22 • 29 | 18 | 20 | 30 | 13 | 42 | 32 | 35 | 40 | 20

Challenge • 23-26 • 8 | 30 | 6 | 42

2.3 • Write the Expression

OUT OF 22

Warm-Up • 1-12 • 12 | 8 | 14 | 6 | 4 | 10 | 20 | 12 | 18 | 13 | 6 | 22

Core • 13-22 • 17 | 17 | 20 | 28 | 49 | 23 | 26 | 27 | 34 | 81

Challenge • 23-26 • 16 | 28 | 30 | 48

2.4 • Two Letters

OUT OF 22

Warm-Up • 1-12 • 5 | 6 | 4 | 8 | 2 | 9 | 7 | 12 | 7 | 10 | 3 | 11

Core • 13-22 • 19 | 24 | 13 | 9 | 9 | 26 | 25 | 25 | 13 | 10

Challenge • 23-26 • 13 | 20 | 12 | 26

Fri • Perimeter as a Rule

OUT OF 8

Warm-Up • 1-4 • 12 | 20 | 16 | 32

Core • 5-8 • 36 | 7 | 44 | 11

Challenge • 9-14 • 26 | 5 | 25 | 38 | 6 | 36

Keep It Level

Take the same amount from both sides and the balance holds.

Warm-Up 12 ITEMS • RECALL

$$1 \quad x + 4 = 11$$

$$2 \quad x + 5 = 12$$

$$3 \quad x + 10 = 15$$

$$4 \quad x + 2 = 9$$

$$5 \quad x + 7 = 7$$

$$6 \quad x + 12 = 20$$

$$7 \quad x + 6 = 13$$

$$8 \quad x + 8 = 15$$

$$9 \quad x + 12 = 18$$

$$10 \quad x + 3 = 11$$

$$11 \quad x + 9 = 9$$

$$12 \quad x + 15 = 25$$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

$$13. \quad x + 17 = 42$$

$$14. \quad x - 5 = 9$$

$$15. \quad x - 23 = 19$$

$$16. \quad x + 38 = 100$$

$$17. \quad x - 14 = 27$$

$$18. \quad x + 29 = 55$$

$$19. \quad x - 8 = 14$$

$$20. \quad x - 35 = 27$$

$$21. \quad x + 56 = 120$$

$$22. \quad x - 19 = 42$$

Multiply and Divide

Divide both sides by the same number and the balance still holds.

Warm-Up 12 ITEMS • RECALL

1 $3x = 21$

2 $6x = 30$

3 $2x = 14$

4 $5x = 25$

5 $x \div 2 = 8$

6 $4x = 20$

7 $4x = 24$

8 $7x = 35$

9 $3x = 18$

10 $9x = 45$

11 $x \div 3 = 7$

12 $5x = 40$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. $8x = 96$

14. $7x = 84$

15. $x \div 5 = 13$

16. $4x = 100$

17. $9x = 108$

18. $6x = 90$

19. $8x = 120$

20. $x \div 6 = 14$

21. $11x = 132$

22. $12x = 156$

Check Your Answer

Put it back in. If both sides match, you are done — no marking needed.

Warm-Up 12 ITEMS • RECALL

1 $x = 7$ in $x + 4$ — the value

2 $x = 5$ in $3x$ — the value

3 $x = 9$ in $x - 2$

4 $x = 4$ in $2x + 1$

5 $x = 6$ in $x \div 2$

6 $x = 10$ in $x + 10$

7 Every answer in the 4 times table is a multiple of what

8 Every answer in the 6 times table is a multiple of what

9 Every answer in the 8 times table is a multiple of what

10 Every answer in the 9 times table is a multiple of what

11 2×5 — does it end in 0 or 5

12 4×5 — does it end in 0 or 5

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

22. Solve then check: $3x = 27$

23. Check $x = 9$ in $3x$ — the value

24. Solve then check: $x + 19 = 31$

25. Check $x = 12$ in $x + 19$

26. Solve: $6x = 72$

27. 3×5 — does it end in 0 or 5

28. 5×5 — does it end in 0 or 5

29. 7×5 — does it end in 0 or 5

30. 9×5 — does it end in 0 or 5

31. Adding two odd numbers gives an odd or an even answer

32. Adding an odd and an even number gives



SHOW THE ROOMS, THE GRID, THE NUMBER LINE



ONE SENTENCE PER MISTAKE

Word to Equation

Turn the situation into an equation, then solve it.

Warm-Up 12 ITEMS • RECALL

1 A number plus 5 is 12.
The number

2 Double a number is 14.
The number

3 A number less 3 is 8. The
number

4 Three lots of a number is
18. The number

5 Half a number is 9. The
number

6 A number plus itself is 16.
The number

7 A number plus 8 is 15.
The number

8 Double a number is 18.
The number

9 A number less 5 is 12.
The number

10 Four lots of a number is
24. The number

11 Half a number is 11. The
number

12 A number plus itself is
24. The number

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Four friends share equally and get 13 each. The total

.....

14. After spending \$18 you have \$27. You started with

.....

15. Five boxes hold 60 altogether. Each box

.....

16. A number times 7 is 91. The number

.....

17. Twice a number plus 3 is 21. The number

.....

18. Five friends share equally and get 14 each. The total

.....

19. After spending \$23 you have \$34. You started with

.....

20. Six boxes hold 90 altogether. Each box

.....

21. A number times 8 is 96. The number

.....

22. Twice a number plus 5 is 27. The number

.....

Mid-Unit Quiz

Eight items across Weeks 1–3. 85% to keep flying.

Warm-Up 4 ITEMS • RECALL

1 6, 11, 16, 21 — next
_____2 $n = 4$. Value of $3n$
_____3 4, 9, 14, 19 — next
_____4 $n = 5$. Value of $4n$

Core GRADED • SHOW YOUR WORKING

STOP RULE This one is a test. Do **every** problem — no stopping early.5. Machine $\times 4 + 2$. In 9
.....6. That machine gives 30 — what went in
.....7. $n = 7$. Value of $3n - 4$
.....8. $x + 18 = 45$
.....9. $7x = 63$
.....10. Machine $\times 5 + 3$. In 8
.....11. That machine gives 28 — what went in
.....12. $n = 8$. Value of $4n - 6$
.....13. $x + 26 = 51$
.....14. $8x = 72$
.....

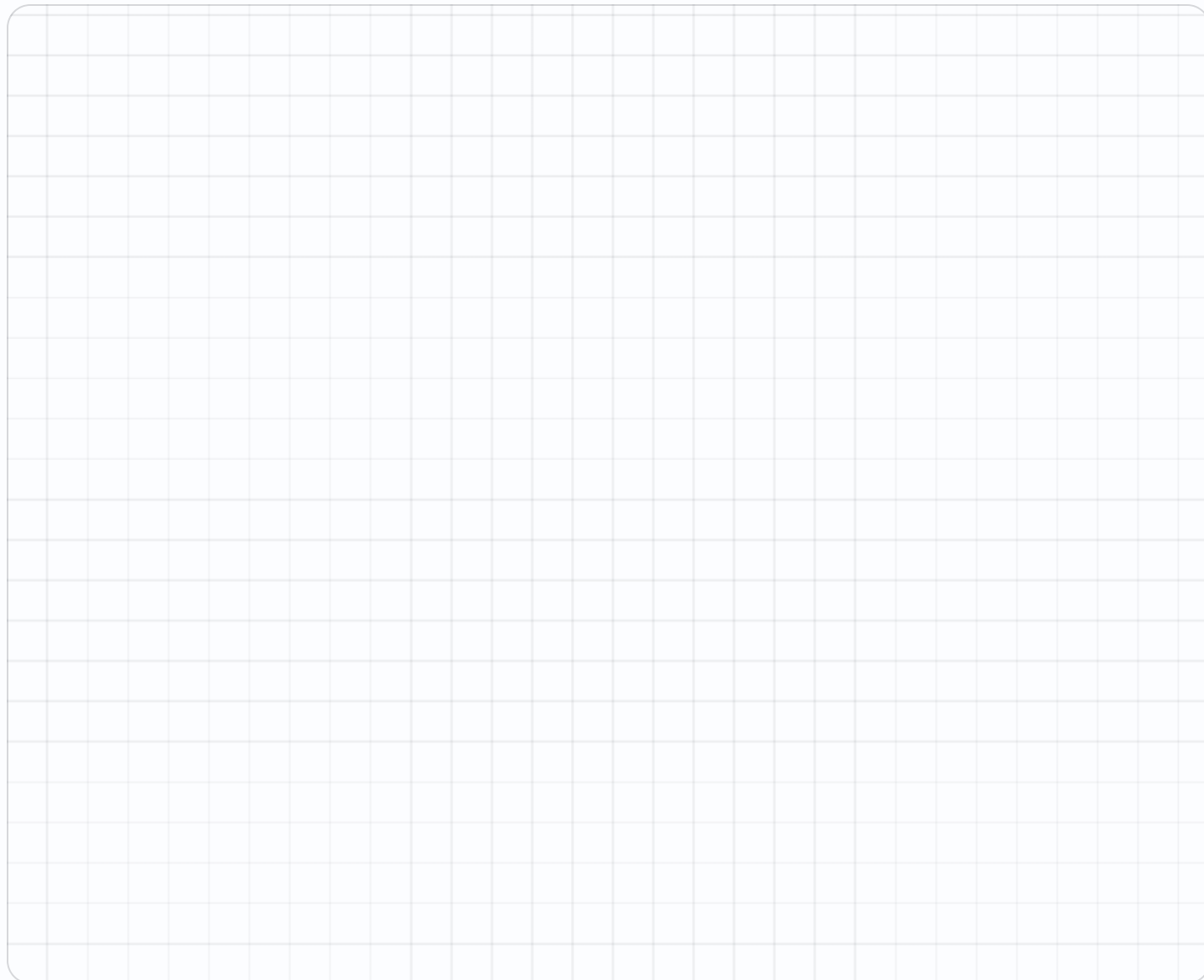
★ Challenge NOT GRADED • REASON IT OUT

15. A pattern starts at 3 and adds 4 — its tenth number

.....

16. A pattern starts at 5 and adds 6 — its tenth number

.....

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Week 3 Answers

Mark the reasoning, not the handwriting.

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

3.1 • Keep It Level

OUT OF 22

Warm-Up • 1-12 • 7 | 7 | 5 | 7 | 0 | 8 | 7 | 7 | 6 | 8 | 0 | 10

Core • 13-22 • 25 | 14 | 42 | 62 | 41 | 26 | 22 | 62 | 64 | 61

Challenge • 23-26 • 244 | 200 | 312 | 300

3.2 • Multiply and Divide

OUT OF 22

Warm-Up • 1-12 • 7 | 5 | 7 | 5 | 16 | 5 | 6 | 5 | 6 | 5 | 21 | 8

Core • 13-22 • 12 | 12 | 65 | 25 | 12 | 15 | 15 | 84 | 12 | 13

Challenge • 23-26 • 24 | 147 | 25 | 171

3.3 • Check Your Answer

OUT OF 23

Warm-Up • 1-12 • 11 | 15 | 7 | 9 | 3 | 20 | 4 | 6 | 8 | 9 | 0 | 0

Core • 22-32 • 9 | 27 | 12 | 31 | 12 | 5 | 5 | 5 | 5 | even | odd

Challenge • 42-46 • 7 | 19 | square | 7 | 25

3.4 • Word to Equation

OUT OF 22

Warm-Up • 1-12 • 7 | 7 | 11 | 6 | 18 | 8 | 7 | 9 | 17 | 6 | 22 | 12

Core • 13-22 • 52 | 45 | 12 | 13 | 9 | 70 | 57 | 15 | 12 | 11

Challenge • 23-26 • 11 | 12 | 10 | 13

Fri • Mid-Unit Quiz

OUT OF 14

Warm-Up • 1-4 • 26 | 12 | 24 | 20

Core • 5-14 • 38 | 7 | 17 | 27 | 9 | 43 | 5 | 26 | 25 | 9

Challenge • 15-16 • 39 | 59

Design the Trail

Ten stops, one per mission plus two of your own. Map it first.

Warm-Up 9 ITEMS • RECALL

1 Stops on the trail

2 Missions in the year

3 Stops of your own choosing

4 3 minutes per stop, 10 stops — minutes

5 Half of 10 stops

6 Two stops per room — rooms needed

7 4 minutes per stop, 10 stops — minutes

8 Half of 8 missions

9 Two stops per station — stations needed

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

10. Trail rule $\times 2 + 3$ — stop 6 gives

.....

11. That rule — stop 10 gives

.....

12. 10 stops at 4 minutes each — minutes

.....

13. If 3 stops are outdoors, the percent indoors

.....

14. Stops numbered 1 to 10 — their total

.....

15. Trail rule $\times 3 + 2$ — stop 5 gives

.....

16. 10 stops at 5 minutes each — minutes

.....

17. If 4 stops are outdoors, the percent indoors

.....

18. Stops numbered 1 to 12 — their total

.....



Working space



Error journal

Write the Problems

Every stop gets a problem and a hidden answer key in your handwriting.

Warm-Up 8 ITEMS • RECALL

1 23×14

2 $372 \div 3$

3 $2 + 3 \times 4$

4 Perimeter of a 4 by 3 rectangle

5 34×12

6 $484 \div 4$

7 $3 + 4 \times 5$

8 Perimeter of a 5 by 4 rectangle

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

9. $x + 17 = 42$

10. Area 48, one side 6 — the other
_____11. GCF of 24 and 36
_____12. Round 4,829 to the nearest thousand

13. $x + 24 = 51$

14. Area 72, one side 8 — the other
_____15. GCF of 18 and 24
_____16. Round 5,632 to the nearest thousand

 Challenge NOT GRADED • REASON IT OUT

17. $\frac{1}{2} + \frac{1}{4}$ — type as a/b

18. $\$20 - \13.68

19. Mean of 4, 7, 7, 9, 13

20. $2/3 + 3/4$ — type as a/b

21. A machine $\times 4 + 3$ — in 10 gives

22. $\frac{1}{2} + \frac{1}{3}$ — type as a/b

23. \$20 - \$11.35

24. Mean of 6, 8, 8, 10, 13

25. $\frac{3}{4} + \frac{2}{3}$ — type as a/b

26. A machine $\times 5 + 2$ — in 12 gives

 Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE

Error journal ONE SENTENCE PER MISTAKE

Mom Walks It

You mark her work. Any ambiguous question gets rewritten on the spot.

Warm-Up 12 ITEMS • RECALL

1 10 stops, 8 correct — the percent

2 10 stops, 9 correct — the percent

3 10 stops, 10 correct — the percent

4 10 minus 7

5 Half of 10 stops

6 10 stops, 5 correct — the percent

7 10 stops, 7 correct — the percent

8 10 stops, 6 correct — the percent

9 10 stops, 3 correct — the percent

10 10 minus 4

11 A quarter of 20 stops

12 10 stops, 2 correct — the percent

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. She takes 4 minutes a stop — total minutes
.....

14. Two stops rewritten out of 10 — the percent
.....

15. 85% of 10 stops — how many
.....

16. If 9 of 10 are solvable, the percent that are not
.....

17. She takes 5 minutes a stop — total minutes
.....

18. Three stops rewritten out of 10 — the percent
.....

19. 90% of 10 stops — how many
.....

20. If 8 of 10 are solvable, the percent that are not
.....

Year-End Sweep

All eight error journals. Name the one habit that fixed itself.

Warm-Up 8 ITEMS • RECALL

1 7×8

2 $84 \div 12$

3 $2 + 3 \times 4$

4 $1/2$ of 20

5 6×9

6 $96 \div 12$

7 $3 + 4 \times 5$

8 $1/3$ of 30

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

9. 38×24

10. $618 \div 6$

11. $3x = 21$

12. 47×23

13. $728 \div 8$

14. $4x = 32$

Mission 08 Test

Twelve items plus the completed trail. Final trophy band awarded.

Warm-Up 4 ITEMS • RECALL

1 3, 6, 9, 12 — next
_____2 $n = 5$. Value of $2n$
_____3 5, 10, 15, 20 — next
_____4 $n = 6$. Value of $3n$

Core GRADED • SHOW YOUR WORKING

STOP RULE This one is a test. Do **every** problem — no stopping early.5. 2, 6, 18, 54 — next
.....6. A pattern from 5 adding 6 — its twelfth number
.....7. Machine $\times 5 + 3$. In 8
.....8. Machine $\times 6 - 4$ gives 32 — what went in
.....9. $n = 9$. Value of $4n + 7$
.....10. $a = 4$, $b = 6$. Value of $3a + 2b$
.....11. $x - 26 = 19$
.....12. $9x = 108$
.....13. 3, 9, 27, 81 — next
.....14. Machine $\times 6 + 2$. In 7
.....15. Machine $\times 5 - 3$ gives 32 — what went in
.....16. $n = 8$. Value of $5n + 6$
.....17. $a = 5$, $b = 7$. Value of $3a + 2b$
.....

Challenge NOT GRADED • REASON IT OUT

20. $2x + 5 = 19$

21. A machine turns 5 into 23 and 8 into 35 — in 10 gives

22. A pattern from 4 adding 7 — its tenth number

23. $3x + 4 = 25$

24. A machine turns 4 into 19 and 7 into 34 — in 10 gives

 Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE

Error journal ONE SENTENCE PER MISTAKE

Week 4 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

4.1 • Design the Trail

OUT OF 18

Warm-Up • 1-9 • 10 | 8 | 2 | 30 | 5 | 5 | 40 | 4 | 5

Core • 10-18 • 15 | 23 | 40 | 70 | 55 | 17 | 50 | 60 | 78

Challenge • 19-22 • 37 | 120 | 46 | 150

4.2 • Write the Problems

OUT OF 16

Warm-Up • 1-8 • 322 | 124 | 14 | 14 | 408 | 121 | 23 | 18

Core • 9-16 • 25 | 8 | 12 | 5000 | 27 | 9 | 6 | 6000

Challenge • 17-26 • $\frac{3}{4}$ | 6.32 | 8 | $\frac{17}{12}$ | 43 | $\frac{5}{6}$ | 8.65 | 9 | $\frac{17}{12}$ | 62

4.3 • Mom Walks It

OUT OF 20

Warm-Up • 1-12 • 80 | 90 | 100 | 3 | 5 | 50 | 70 | 60 | 30 | 6 | 5 | 20

Core • 13-20 • 40 | 20 | 8.5 | 10 | 50 | 30 | 9 | 20

Challenge • 21-26 • 4 | 80 | 102 | 5 | 80 | 127.5

Thu • Year-End Sweep

OUT OF 22

Warm-Up • 1-12 • 56 | 7 | 14 | 10 | 0.5 | 90 | 54 | 8 | 23 | 10 | 1 | 180

Core • 13-22 • 912 | 103 | $\frac{11}{12}$ | 9 | 7 | 1081 | 91 | $\frac{11}{12}$ | 9 | 8

Challenge • 23-26 • 25 | 7 | 15 | 7

Fri • Mission 08 Test

OUT OF 20

Warm-Up • 1-4 • 15 | 10 | 25 | 18

Core • 5-20 • 162 | 71 | 43 | 6 | 43 | 24 | 45 | 12 | 243 | 67 | 44 | 7 | 46 |
29 | 55 | 12

Challenge • 21-24 • 7 | 43 | 7 | 49